

Muhammad Nazil Haziq bin Mohd Nizar

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CAREER OBJECTIVE

Motivated Computer Engineering student at UTP with hands-on experience in full-stack development, AI-powered applications, and mobile development. Seeking a software engineering or AI engineering internship from September 2026 – April 2027 to contribute strong skills in Flutter, Python, FastAPI, React, and machine learning to real-world software solutions.

EDUCATION

Universiti Teknologi PETRONAS (UTP)

Bachelor of Computer Engineering with Honours

May 2024 – April 2028 (Expected)

CGPA: 3.34 / 4.0

- Dean's List Award — Semester 3 & 6
- Relevant coursework: Data Structures & Algorithms, Software Engineering, Data Analytics, Object-Oriented Programming, Structured Programming & Interfacing, Microprocessor & Computer Architecture, Digital System Design

TECHNICAL SKILLS

Languages: Python, C, C++, JavaScript, Dart (Flutter), ARM Assembly, MATLAB

Frameworks & Libraries: FastAPI, Flutter, React, Pydantic, Firebase, pandas, NumPy, Scikit-learn

AI & Backend: RAG pipelines, Vector embeddings, pgvector, PostgreSQL, REST APIs, JWT authentication, Async Python

Embedded & Hardware: ESP32, Arduino, STM32 (NUCLEO-F401RE), I2C, ADC, GPIO

Tools & Platforms: Git, GitHub, Firebase Cloud Firestore, VS Code, Simulink, Power BI

Concepts: OOP, Cloud Synchronisation / Synchronization, Real-time Data Streaming, SDLC

PROJECTS

AI Maintenance Copilot

June 2026

FastAPI, PostgreSQL, pgvector, React, OpenAI Embeddings, RAG, JWT Auth

- Built a full-stack AI assistant for industrial engineers to query PDF maintenance manuals using Retrieval-Augmented Generation (RAG), engineers ask questions in natural language and receive cited answers sourced from the correct manual page
- Designed and implemented the backend from scratch: FastAPI with async endpoints, PostgreSQL database with 5 normalised tables, pgvector for semantic chunk storage, and JWT-based authentication with bcrypt password hashing
- Engineered the full RAG pipeline: PDF ingestion, sliding-window chunking, OpenAI embedding generation, and pgvector cosine similarity search, enabling engineers to retrieve accurate, cited answers from 300-page manuals in under 3 seconds

IngatKu | AI Memory Companion for Malaysian Elderly (ElectroHack 2.0 Hackathon)

May 2026

Flutter, Firebase Cloud Firestore, Gemini API, Dart

- Led a 5-person team as Project Lead & Frontend Developer to build a cross-platform mobile app that uses Gemini 2.5 Flash AI to automatically read notification messages, extract appointment and medication details, and create reminders
- Implemented real-time Firebase Firestore sync, medication streak tracking with scheduled alarms, emergency help button, and Cabby transport booking integration

Smart Parking System for Micromobility (Dockify)

Sep 2025 – April 2026

Flutter, C++, ESP32, Arduino

- Led development of a cross-platform mobile app, ESP32 firmware in C++, and hardware circuit design as part of a 5-person team. Prototype delivered within RM 105 budget, rated 4.17/5 for parking efficiency improvement by students
- Implemented QR-based slot detection and automated locking system, addressing a campus micromobility problem where 51% of users felt unsafe with existing parking infrastructure

EV Digital Twin System | NVIDIA Omniverse

Jan 2026

Flutter, Firebase Cloud Firestore, Python, Scikit-learn

- Built a Flutter mobile dashboard for real-time EV monitoring, displaying 5 live telemetry parameters including battery state of health, predicted driving range, and safety alerts via Firebase Cloud Firestore with 1-second update intervals
- Integrated Firebase as the cloud sync layer connecting a Random Forest prediction pipeline to the mobile app; collaborated in a 4-person team owning the mobile UI, cloud integration, and contributing to AI model development

EXTRA-CURRICULAR ACTIVITIES

Logistics Volunteer | CAVE 2026 (Campus Venture)

May 2026

Universiti Teknologi PETRONAS

- Supported logistics operations for one of UTP's largest annual events, coordinating across ~100 participating clubs showcasing to incoming students
- Managed stage area setup and on-ground coordination during event day, responding to real-time changes in layout and programme flow

Event Management Committee | EE and COE First Year Bootcamp

Jan 2026

E3S2 — Electrical & Electronics Engineering Students Society, UTP

- Coordinated event programmes and logistics for a leadership bootcamp involving ~50 incoming EE and COE first-year students, managing activity flow, F&B distribution, and on-ground operations
- Contributed to pre-event planning including programme scheduling, resource preparation, and logistical coordination across the organising committee

Event Management Committee | Palestine Solidarity Run

Sep 2025

UTP Typhoon Athletics Club

- Assisted in coordinating event flow and on-ground logistics for a campus-wide charity run with 100–150 participants, organised by UTP Typhoon athletics club
- Stationed at a designated checkpoint during the run to monitor participant safety and ensure correct route adherence

Logistics and Public Relations Committee | MentorLink

Dec 2025

Community Service Programme, UTP

- Coordinated logistics and participant outreach for a campus mentorship programme involving 37 participants from Rumah Anak Yatim Dahikmah, liaising between committees and external stakeholders
- Handled end-to-end logistics including transport booking, venue arrangement, catering, and materials procurement prior to the event

TRAINING AND CERTIFICATIONS

Artificial Intelligence Fundamentals

IBM SkillsBuild

MATLAB Fundamentals

MathWorks

Simulink Fundamentals

MathWorks

Python Fundamentals for Beginners

Great Learning

ADDITIONAL

Languages: English (Professional), Malay (Native) **Interests:** Sports (Running, Football), Technology & Finance News

REFERENCE

Ir Dr Ashvaany Egambaram

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